

1.8 Reaction Kinetics

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.8 Reaction Kinetics
Difficulty	Medium

Time allowed: 60

Score: /49

Percentage: /100

Question 1a

Fig. 1.1 below shows, for a given temperature T , a Boltzmann distribution of the kinetic energy of the molecules of a mixture of two gases that will react together.

The activation energy for the reaction, E_a , is marked.

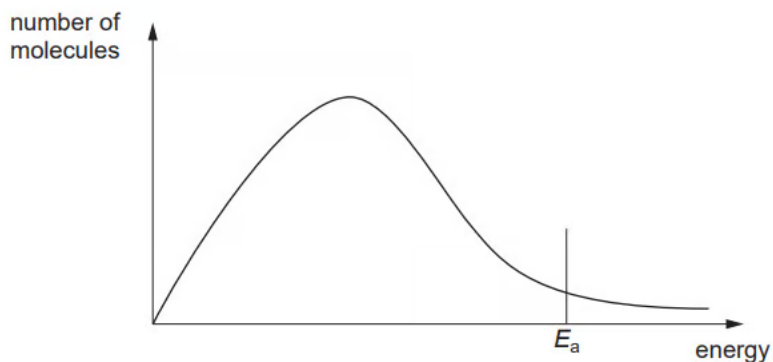


Fig. 1.1

On Fig. 1.1 above,

- i)
draw a new distribution curve, clearly labelled T_2 , for the same mixture of gases at a lower temperature, T_2 .

[2]

- ii)
mark clearly, as **H**, the position of the activation energy of the reaction at the lower temperature, T_2 .

[1]

[3 marks]

Question 1b

Explain the meaning of the term **activation energy**.

[2 marks]

Question 1c

i)

On the energy axis in Fig. 1.1, mark the position, clearly labelled **C**, of the activation energy of the reaction when a catalyst is used.

[1]

ii)

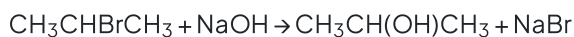
Use your answer to **(i)** to explain how the use of a catalyst results in reactions occurring at a faster rate.

[1]

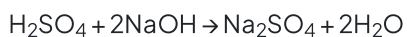
[2 marks]

Question 1d

Two reactions involving aqueous NaOH are given below.



reaction 1



reaction 2

The reagents in **reaction 1** must be heated together for some time for the reaction to occur.

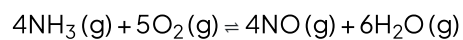
Whereas, **reaction 2** is almost instantaneous at room temperature.

Suggest brief explanations why the rates of these two reactions are very different.

[4 marks]

Question 2a

A platinum-rhodium catalyst used in the reaction forming nitrogen(II) oxide from ammonia.



i)

State what is meant by the term heterogeneous catalyst.

[1]

ii)

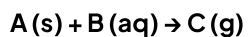
Explain why a catalyst has no effect on the yield of the products in the reaction.

[1]

[2 marks]

Question 2b

A group of students were completing a practical, investigating the factors which affect the rate of the following chemical reaction.



The students collected the gas produced and plotted the graph shown in Fig. 1.1.

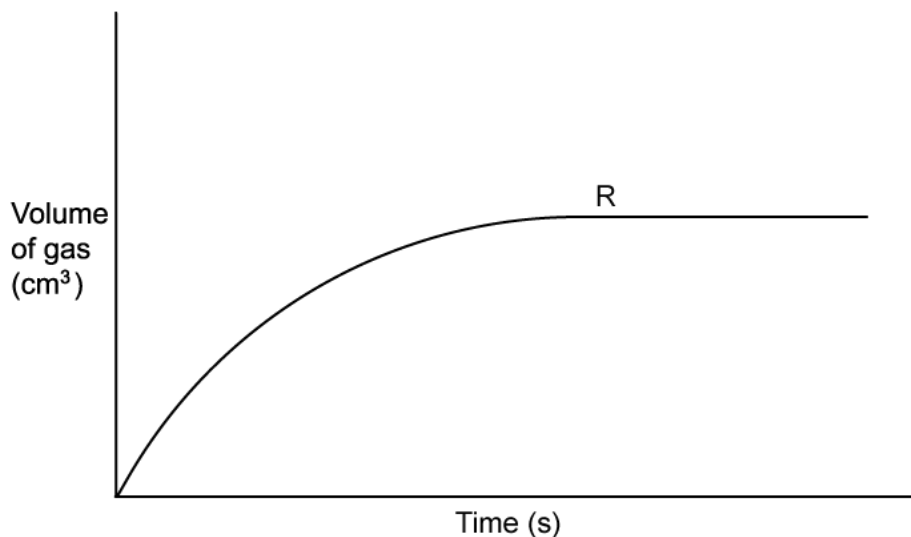


Fig. 1.1

- i)
State and explain what the letter **R** represents on the student's graph.

[1]

- ii)
In the original reaction above, the students used 0.5 g of **A** and 50 cm³ of 1.0 mol dm⁻³ of **B**. Sketch a curve on the graph in Fig 1.1 to show how the total volume of gas collected would change if the students still used 0.5 g of **A**, but used 50 cm³ of 2.0 mol dm⁻³ of **B**.

[2]

[3 marks]

Question 2c

Explain why the gradient of the curve in part (b) decreases as the time of the reaction progresses.

[2 marks]

Question 2d

Another way to increase the rate of reaction is to increase the temperature.

Explain why a small increase in temperature has a large effect on the initial rate of a chemical reaction.

[2 marks]

Question 3a

This question is about how different factors affect the rate of a chemical reaction.

A reaction between reactant molecules can only occur when collisions between the reactant molecules are effective.

i)

Explain what is meant by the term effective collision.

[1]

ii)

State **two** factors that could cause an ineffective collision.

[2]

[3 marks]

Question 3b

A student made the following statement about factors that affect the rate of a chemical reaction.

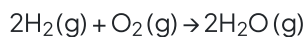
'The rate of a reaction increases with temperature as there are more collisions'

Discuss why this statement is only true to an extent and what other factors need to be considered to fully explain how the rate of a reaction is affected by temperature.

[5 marks]

Question 3c

Hydrogen and oxygen can react to form water in an exothermic reaction.



The rate of this reaction depends on the pressure as well as the temperature.

Describe and explain the effect of increasing the pressure on this reaction.

[2 marks]

Question 3d

If the rate of reaction increases suddenly, an explosion can occur.

A highly exothermic reaction, such as the reaction in **(c)**, is at more risk of exploding than a less exothermic reaction.

Suggest why.

[2 marks]

Question 4a

A series of experiments was carried out to investigate the factors which affect the rate of reaction between calcium carbonate and dilute hydrochloric acid.



- 50.0 cm³ of hydrochloric acid was added to 10 g of small pieces of calcium carbonate (an excess) in a conical flask placed on an electronic balance.
- The loss in mass of the flask and its contents was recorded every 30 seconds for 10 minutes.
- The experiment was repeated using a different concentration of hydrochloric acid or a different temperature.
- The results are shown in Table 1.1

Table 1.1

Experiment	Concentration of hydrochloric acid / mol dm ⁻³	Temperature / °C
1	0.50	20
2	0.50	60
3	1.00	20

The results of Experiment 1 are shown in Fig. 1.1.

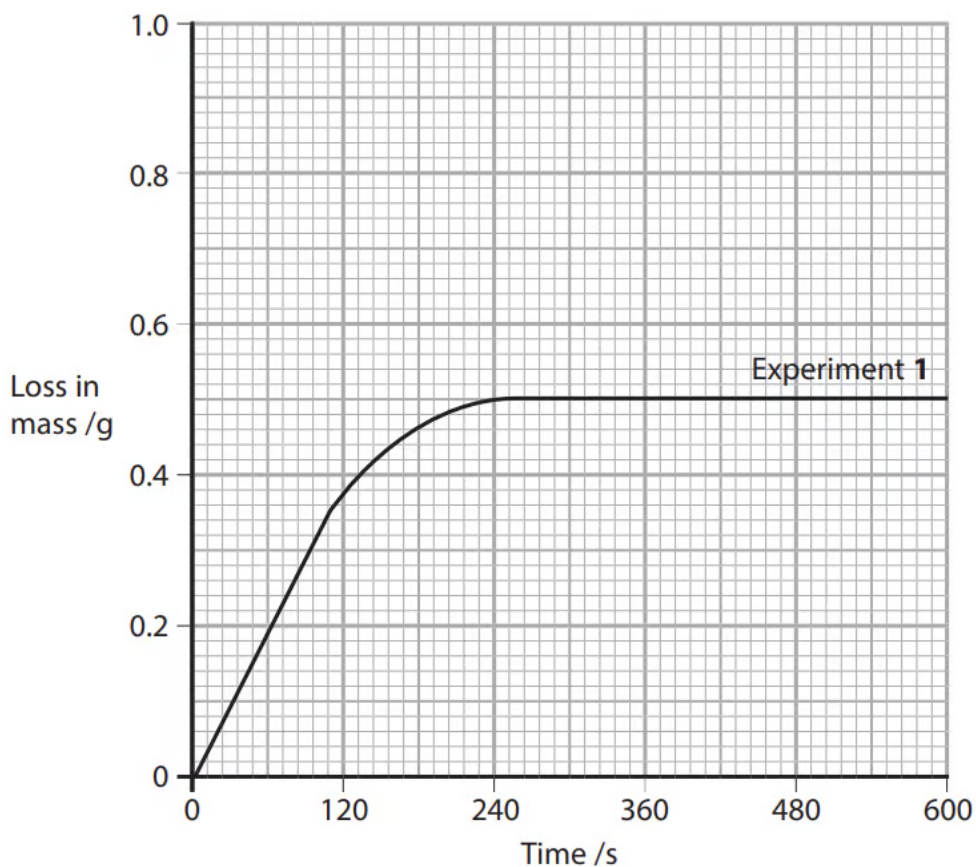


Fig. 1.1

Draw curves on the graph in Fig 1.1 to show the results you would expect for Experiments **2** and **3**.

Label the curves **2** and **3**.

[2 marks]

Question 4b

Determine the initial rate of reaction for Experiment **1**.

You must show your working on the graph in Fig 1.2.

Include units in your answer.

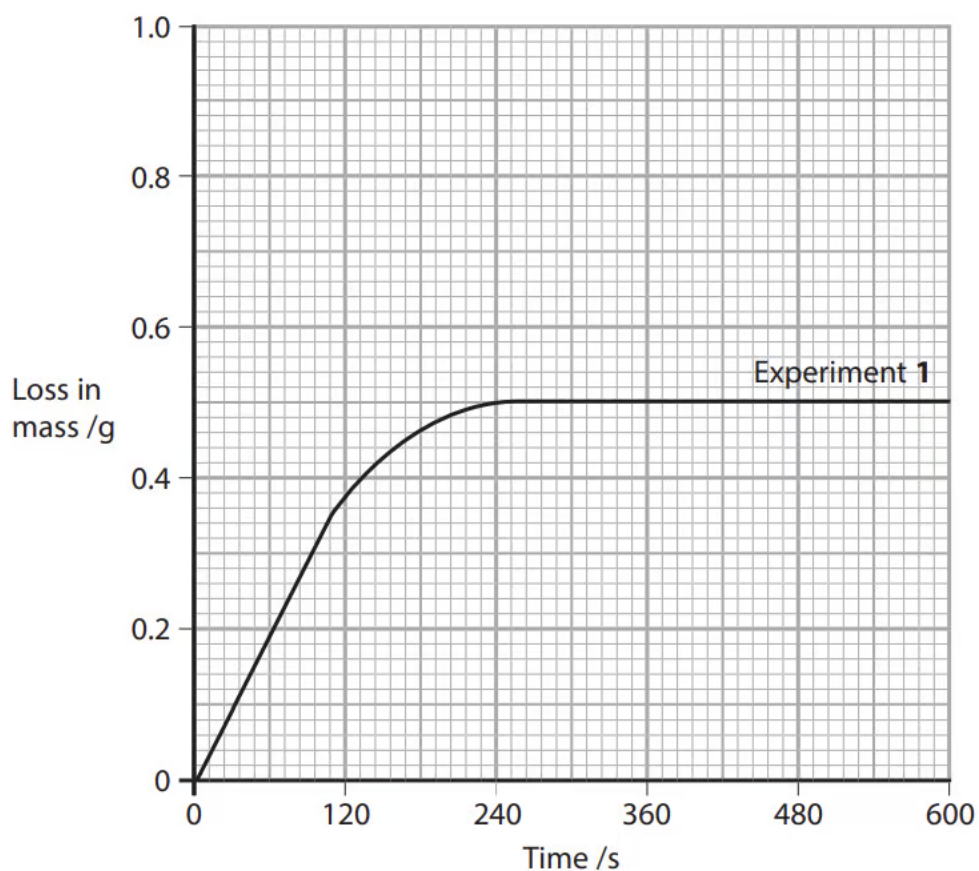


Fig 1.2

Initial rate of reaction

[3 marks]

Question 4c

Calculate the average rate of reaction over the first 240 seconds of Experiment 1.

Average rate of reaction.....

[1 mark]

Question 4d

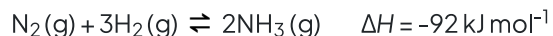
The student repeated each experiment but used 10.0 g of larger sized pieces of calcium carbonate.

Suggest what effect this would have on the rate of reaction. Give your reasoning.

[2 marks]

Question 5a

Ammonia can be produced by the reaction of nitrogen and hydrogen.



The reaction can be catalysed and the activation energy for this catalysed reaction is $+109 \text{ kJ mol}^{-1}$

Complete the reaction pathway diagram in Fig. 1.1 for the uncatalysed **and** the catalysed reaction between nitrogen and hydrogen.

You should label the following:

- products
- the enthalpy change of reaction, ΔH
- the activation energy of the forward, uncatalysed reaction, E_a
- the activation energy of the forward, catalysed reaction, E_c

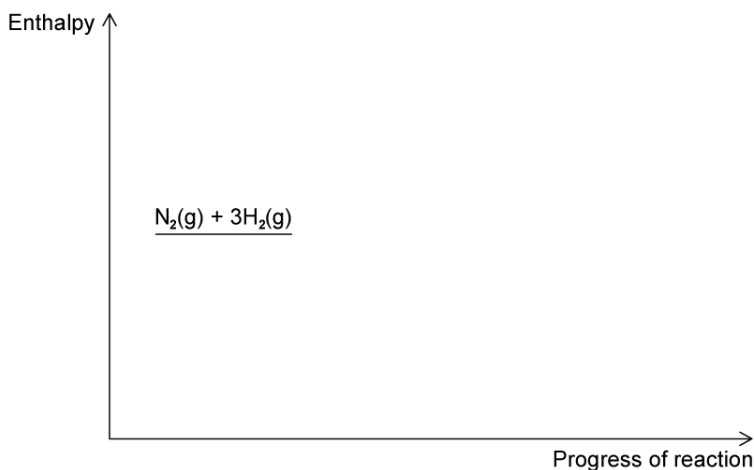


Fig. 1.1

[4 marks]

Question 5b

Calculate the value of the activation energy of the catalysed decomposition of ammonia into nitrogen and hydrogen.

Show your working.

activation energy kJ mol^{-1}

[1 mark]

Question 5c

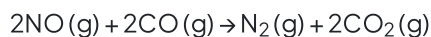
Catalysts, such as iron used in the production of ammonia, increase the rate of reaction.

Explain why. Use a labelled Boltzmann distribution to explain your answer.

[1 mark]

Question 5d

Platinum is used as a catalyst in catalytic converters which are fitted to vehicle exhaust systems to remove nitrogen oxide from the exhaust gases.



i)

State the type of catalyst that platinum is in a catalytic converter

[1]

ii)

Explain, using oxidation numbers, if nitrogen is being oxidised or reduced in this reaction.

[2]

[3 marks]

Model Answers: Medium

1a

a)

i) The new distribution curve for the same mixture of gases at a lower temperature should have the following features:

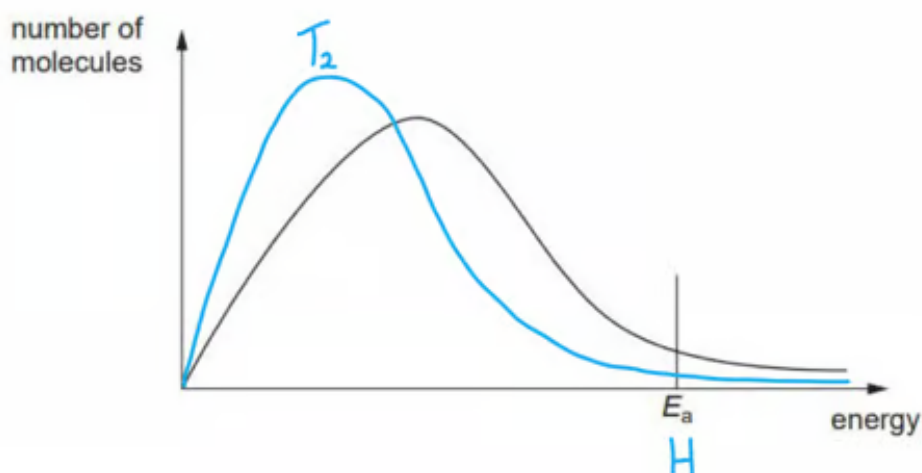
- The new graph has a higher maximum; [1 mark]
- The maximum is to the left of the previous maximum; [1 mark]

ii) The position of **H**:

- **H** is at E_a ; [1 mark]

[Total: 3 marks]

- An example of a correct diagram is:



- It is not uncommon for students to misread this and draw a Boltzmann distribution for a higher temperature, as this is more commonly examined
- The peak should be to the left of the original as the average energy of the molecules is lower

- The peak is higher as more molecules have the average energy
- Check that the 2 curves only cross each other once – if they cross more times than this, you know you have made an error in your curve
- The activation energy does not change with temperature so will be in the same position

1b

b) The meaning of the term **activation energy** is:

- The minimum amount of energy molecules / particles must have
OR
The energy required; [1 mark]
- (In order) for a reaction to take place; [1 mark]

[Total: 2 marks]

- This is a key definition that you need to learn, and an easy 2 marks to pick up
- It is worth making a note of all of the definitions on the specification and taking the time to learn them

1c

c)

i) The position of **C** is

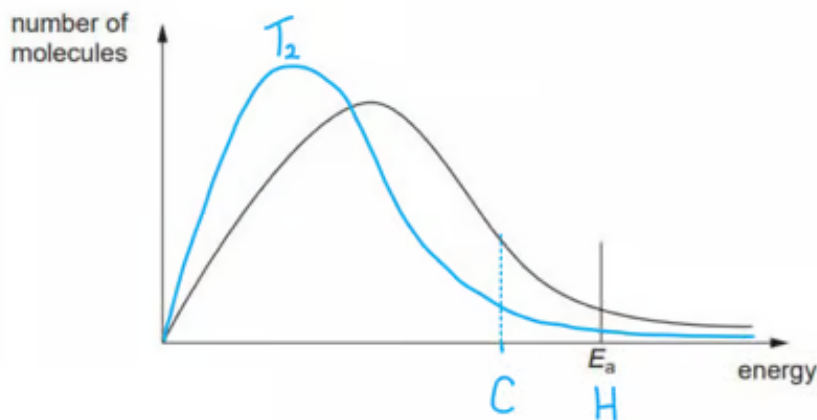
- **C** should be placed to the left of **H**; [1 mark]

ii) The use of a catalyst results in reactions occurring at a faster rate because:

- More molecules / particles now have an energy that exceeds the activation energy / $> E_a$; [1 mark]

[Total: 2 marks]

- An example of a correct diagram is:



- A catalyst provides an alternative reaction mechanism that has a lower activation energy, which is why the position of **C** is at lower energy than **H**
- This means that there are a greater proportion of molecules that have an energy greater than E_a so there will be more effective collisions and, therefore, an increased rate of reaction

1d

d) Explanations of why the rates of these two reactions are very different are:

- **Reaction 1** has greater E_a ; [1 mark]
- Because energy is needed to break covalent bonds; [1 mark]
- (**Reaction 2** has a lower E_a because) the reaction involves ions
OR
(**Reaction 2** has a lower E_a because) the actual reaction is $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$; [1 mark]
- Opposites attract; [1 mark]

[Total: 4 marks]

- The 'suggest' command indicates that this is not something you will have specifically studied, but you will need to apply your knowledge to this situation
- The rate of reaction depends on the activation energy, so you need to look at the reactions to see what would cause one to have a higher activation energy than the other
- The key is the different types of structures of the particle reacting with NaOH
- $\text{CH}_3\text{CHBrCH}_3$ is a simple molecule so will require covalent bonds to be broken when it reacts with NaOH
- H_2SO_4 will completely dissociate in solution so no breaking of ionic bonds will

occur during the reaction, as these have already been overcome

- **Reaction 2** is a neutralisation reaction so the reaction that occurs is
 - $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$
- This occurs readily as the ions are attracted

2a

a)

i) The definition of the term 'heterogeneous catalyst' is:

- The catalyst is in a different phase / state to the reactants; [1 mark]

ii) A catalyst has no effect on the yield of the products in the reaction because:

- (A catalyst) increases the rate of the forward and reverse reactions to the same extent; [1 mark]

[Total: 2 marks]

- A catalyst only speeds up the rate of the reaction therefore the equilibrium is met faster
 - However the same net amount of products are produced, just at a quicker rate

2b

b)

i) The letter **R** on the students' graph represents:

- The rate of reaction at this point is zero

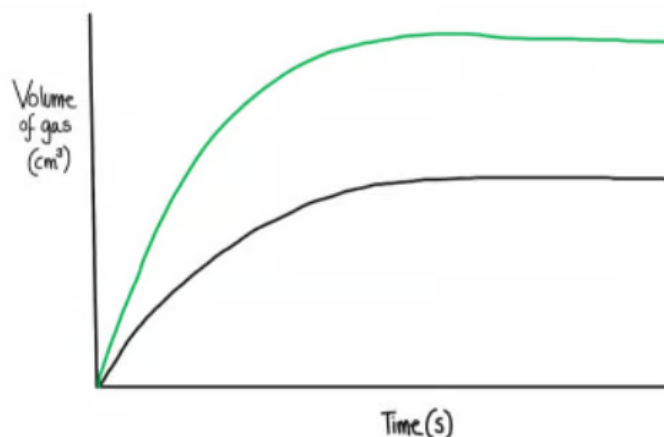
OR

The reaction is complete / has stopped / no more product will be produced

OR

All of the reactants have been used up / reacted; [1 mark]

ii) The sketch of the graph if students used 0.5 g of **A** and 50 cm³ of 2.0 mol dm⁻³ of **B**:



- The curve is steeper than the original curve and still starts at the origin; [1 mark]
- The curve levels / plateaus at double the height of the original curve; [1 mark]

[Total: 3 marks]

- Once all of the reactants have reacted together, no more product can be produced, meaning that the graph will plateau
- The faster the rate of the reaction, the steeper the gradient/slope of the graph
 - In the second experiment, the concentration of **B** is doubled, but **nothing else is changed**
 - So, the reaction happens faster and double the amount of product in this case is formed

2c

c) The gradient of the curve decreases as the time of the reaction progresses because:

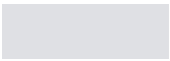
- The concentration of reactants decreases
OR
The reactants start to be used up; [1 mark]
- The rate of the reaction / rate of the formation of the gas decreases
OR
The frequency of successful collisions decreases / falls / is slower; [1 mark]

[Total: 2 marks]

- The faster the rate of the reaction, the steeper the gradient/slope of the graph
 - As reactants start to be used up, the gradient of the graph starts to decrease

- Once all of the reactants have been used up, the graph plateaus and no more product is produced
- The rate of the reaction at this point is zero; the reaction has finished

2d



- The concentration of reactants decreases

OR

The reactants start to be used up; [1 mark]

- The rate of the reaction / rate of the formation of the gas decreases

OR

The frequency of successful collisions decreases / falls / is slower; [1 mark]

[Total: 2 marks]

- The faster the rate of the reaction, the steeper the gradient/slope of the graph
 - As reactants start to be used up, the gradient of the graph starts to decrease
 - Once all of the reactants have been used up, the graph plateaus and no more product is produced
 - The rate of the reaction at this point is zero; the reaction has finished

2d

d) A small increase in temperature has a large effect on the rate of a chemical reaction because:

- Many / significantly more molecules / particles will have energy greater than or equal to the activation energy

OR

A much higher proportion of molecules / particles will have energy greater than or equal to the activation energy; [1 mark]

- The proportion of collisions which are effective / successful will increase

AND

The frequency of successful collisions will increase; [1 mark]

[Total: 2 marks]

- For a chemical reaction to take place, molecules / particles must collide in the correct orientation and with sufficient energy
 - They must have energy greater than or equal to the activation energy
- Increasing the temperature gives the particles more energy, which means that a greater proportion of the particles will have the activation energy
- Particles will also be moving faster and colliding more frequently
- Since the proportion of particles with the activation energy is higher, and the particles are colliding more frequently, the frequency of effective collisions will

increase significantly

- Increasing the temperature has the greatest effect on increasing the rate of the reaction of all of the possible factors

3a

a)

i) The term effective collision means:

- A collision that results in a reaction / products being formed; [1 mark]

ii) **Two** factors that could cause an ineffective collision are:

- The (reactant) particles may not collide in the correct orientation; [1 mark]
- The (reactant) particle may not have an energy greater than the activation energy (of the reaction); [1 mark]

[Total: 3 marks]

- These are some of the key terms for this particular topic that you need to be able to define and understand

3b

b) Discussing the statement about how the rate of reaction is affected by temperature:

- (As temperature increases) there are more collisions (so the statement is true); [1 mark]
- But the rate depends on the frequency of effective collisions (not just the number of collisions); [1 mark]
- As the temperature increases the (average) kinetic energy / speed (of the particles) increases; [1 mark]
- This increases the frequency of collisions; [1 mark]
- There is a higher proportion of collisions that exceeds E_a resulting in more effective collisions; [1 mark]

[Total: 5 marks]

- You should be aware that when explaining increases in the rate of reaction, you should always refer to the increased frequency of collisions, not just stating there

are more collisions

- This is because the rate is a measure of the changes in the amount of reactant/product in a given amount of time – so there need to be more collisions per unit time for this to occur
- When temperature increases, there are two reasons why the rate occurs
 - As the particles are moving quicker due to an increase in kinetic energy, they will collide more frequently
 - A greater proportion of these collisions are effective as more particles have energy that exceeds E_a
- The second factor is the most important in increasing the rate of reaction

3c

c) To describe and explain the effect of increasing the pressure on this reaction:

- The rate increases; [1 mark]
- As there are more molecules / particles in a given volume, so more frequent collisions; [1 mark]

[Total: 2 marks]

- There are keywords that you must use in explanations of rate changes
- As seen in part (b), it is important not just to state 'more collisions', you must give a time factor, such as more collisions per second or more collisions per unit time
- It is also insufficient to state just 'more molecules' – you must explain that there are more molecules in a given volume, or per dm^3 , or other unit of volume
 - **Careful:** Do **not** state there are more molecules in a given **area**
- You could also state that the molecules are closer together for the second mark

3d

d) A highly exothermic reaction is at more risk of exploding than a less exothermic reaction because:

- When the reaction starts the temperature / kinetic energy of the (reactant) particles / speed of the (reactant) particles increases (more than in a less exothermic reaction); [1 mark]
- (This means that) **many** / **many** more particles have an energy greater than E_a , which increases the rate of reaction (more rapidly than in a less exothermic

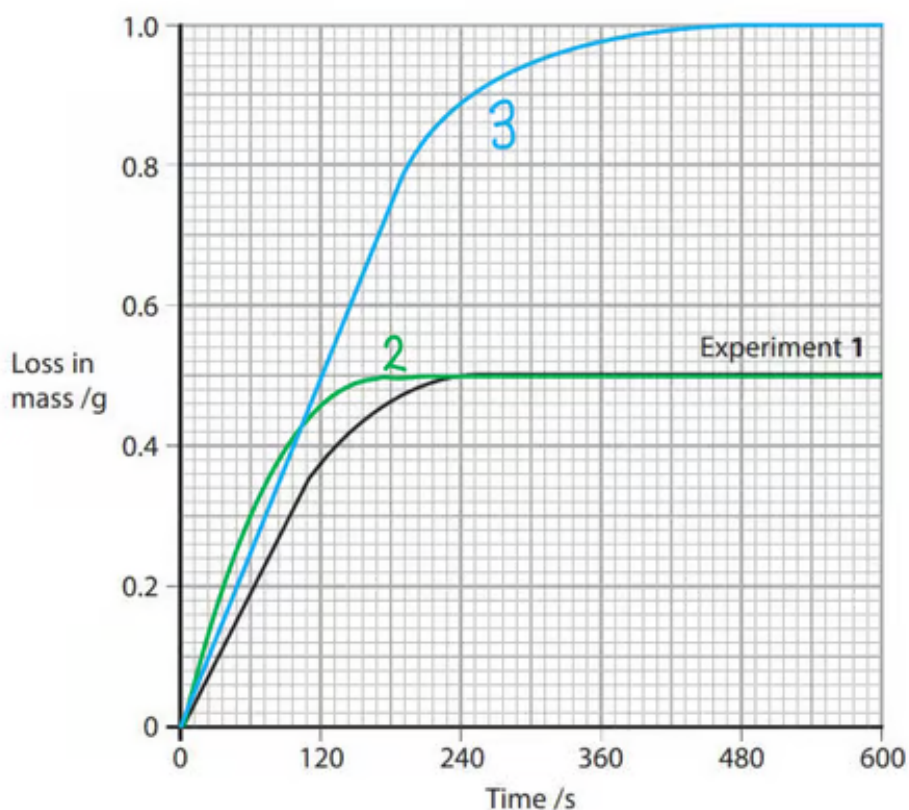
reaction); [1 mark]

[Total: 2 marks]

- The 'suggest' command word indicates that you may not have specifically studied this but you need to apply your knowledge to answer it
- The question implies that as a highly exothermic reaction is more at risk of exploding, then it is more likely that its rate of reaction will increase suddenly
- You need to explain why the rate of reaction of a highly exothermic reaction will increase more suddenly than the rate of a reaction of a less exothermic reaction
- **Remember:** The more molecules with an energy exceeding E_a , the quicker the rate of reaction, so for a reaction to have a sudden increase in the rate of reaction, there must be a sudden increase in the energy of the particles
- Highly exothermic reactions will release more energy than less exothermic reactions
 - After the reaction starts, the particles in the highly exothermic reaction will suddenly gain more energy, causing the rate of reaction to increase as more particles have an energy $> E_a$

4a

a) The graph with curves for experiments **2** and **3** is:



- Experiment 2: Curve steeper than Experiment 1

AND

Curve finishes at the same mass loss as Experiment 1 / 0.5 g; [1 mark]

- Experiment 3: Curve steeper than Experiment 1

AND

Curve finishes at a mass loss of 1.0 g; [1 mark]

[Total: 2 marks]

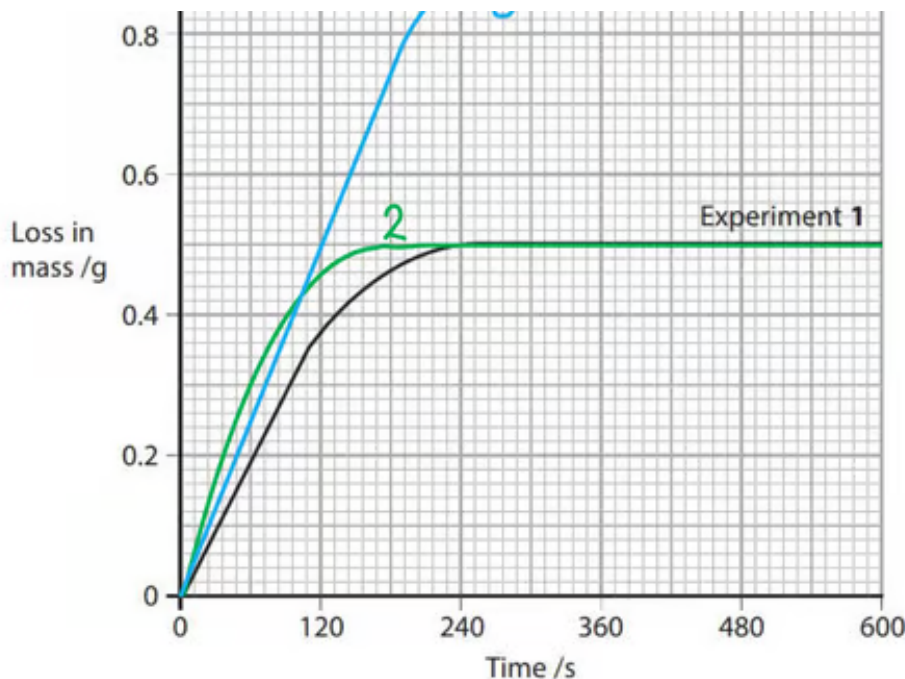
- This question requires you to compare pairs of experiments, identify the change and consider the impact it will have
- Experiments 1 and 2:
 - The temperature of the reaction has increased
 - This gives the particles more energy leading to more frequent successful collisions and a faster rate of reaction (steeper line)
 - The amount of reactants have not changed so the mass loss will be the same
- Experiments 1 and 3:
 - The concentration of HCl has increased
 - This means that there are more acid particles in the same volume leading to more frequent successful collisions and a faster rate of reaction (steeper line)

o HCl is the limiting reagent

■ 10 g of calcium carbonate = $\frac{10}{100} = 0.1$ moles

■ 50cm³ of 0.50 mol dm⁻³ HCl = $0.50 \times \frac{50}{1000} = 0.025$ moles

■ 50cm³ of 1.00 mol dm⁻³ HCl = $1.00 \times \frac{50}{1000} = 0.05$ moles



- Experiment **2**: Curve steeper than Experiment **1**
AND
Curve finishes at the same mass loss as Experiment **1** / 0.5 g; [1 mark]
- Experiment **3**: Curve steeper than Experiment **1**
AND
Curve finishes at a mass loss of 1.0 g; [1 mark]

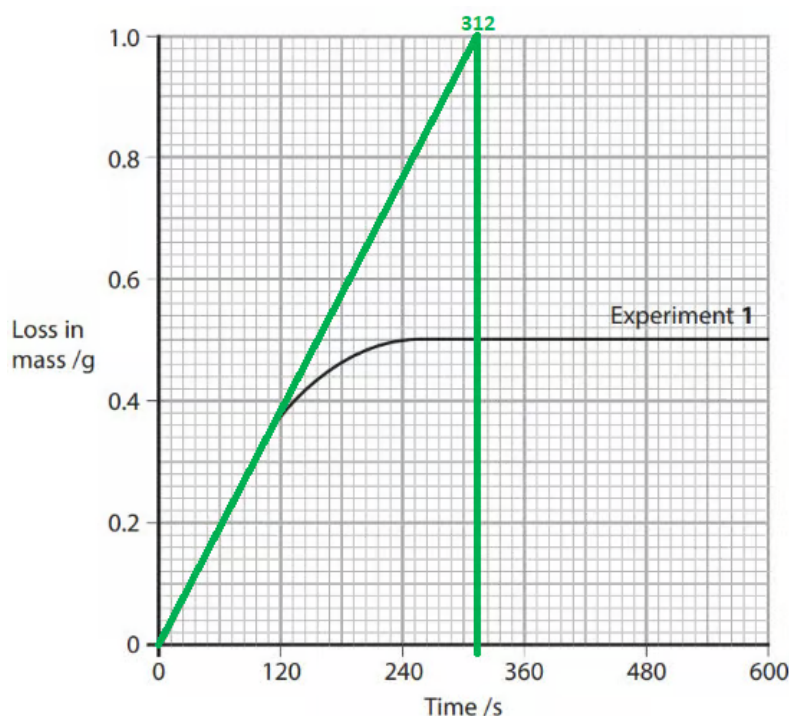
[Total: 2 marks]

- This question requires you to compare pairs of experiments, identify the change and consider the impact it will have
- Experiments **1** and **2**:
 - The temperature of the reaction has increased
 - This gives the particles more energy leading to more frequent successful collisions and a faster rate of reaction (steeper line)
 - The amount of reactants have not changed so the mass loss will be the same
- Experiments **1** and **3**:
 - The concentration of HCl has increased
 - This means that there are more acid particles in the same volume leading to more frequent successful collisions and a faster rate of reaction (steeper line)
 - HCl is the limiting reagent
 - 10 g of calcium carbonate = $\frac{10}{100} = 0.1$ moles

- 50cm^3 of 0.50 mol dm^{-3} HCl = $0.50 \times \frac{50}{1000} = 0.025$ moles
- 50cm^3 of 1.00 mol dm^{-3} HCl = $1.00 \times \frac{50}{1000} = 0.05$ moles
- Since the concentration of HCl has doubled, the mass loss will also double from 0.5 g to 1.0 g

4b

b) The initial rate of reaction for Experiment 1 is:



- Tangent drawn to the curve from time = 0
AND
Tangent extends to at least 120 s; [1 mark]
- Gradient calculation = $\frac{1.0}{312} = 3.21 \times 10^{-3}$
OR
Answers in the range of 3.13×10^{-3} to 3.53×10^{-3} ; [1 mark]
- Gradient units = g s^{-1}
OR
 g / s ; [1 mark]

[Total: 3 marks]

- The question says to show your working on the graph, which means that there is a

mark for drawing the tangent

- When drawing tangents, the line should be extended as far as is convenient for you to perform the calculations
- Extending the tangent in this way decreases the amount of uncertainty
- **Remember:** Gradient = $\frac{\text{change in y - axis}}{\text{change in x - axis}}$
 - **Careful:** Each small square on the x-axis is worth 12 s
 - You can also deduce the units using the gradient equation by substituting the appropriate units for each axis
- If you didn't draw a tangent but used the straight line part of the graph from t = 0 sec to 108 sec where the change in mass is 0.35 g, you would score 2 marks
 - This would give a rate of $3.24 \times 10^{-3} \text{ g / s}$

4c

c) The average rate of reaction over the first 240 seconds of Experiment 1 is:

- $0.0021 / 0.00208 / 0.002083$
- AND**
- g s^{-1} ;

[Total: 1 mark]

- The average rate is calculated by the total mass lost in $240 \text{ s} \div 240$
- Average rate = $0.5 \div 240 = 0.0021 \text{ g s}^{-1}$
- You must give a unit in your answer, however, you would get error carried forward from part (b)

4d

d) The effect of using larger sized pieces would have on the rate of reaction is:

- The rate would decrease; [1 mark]
- The surface area is smaller, so less frequent (effective) collisions (between reactant particles); [1 mark]

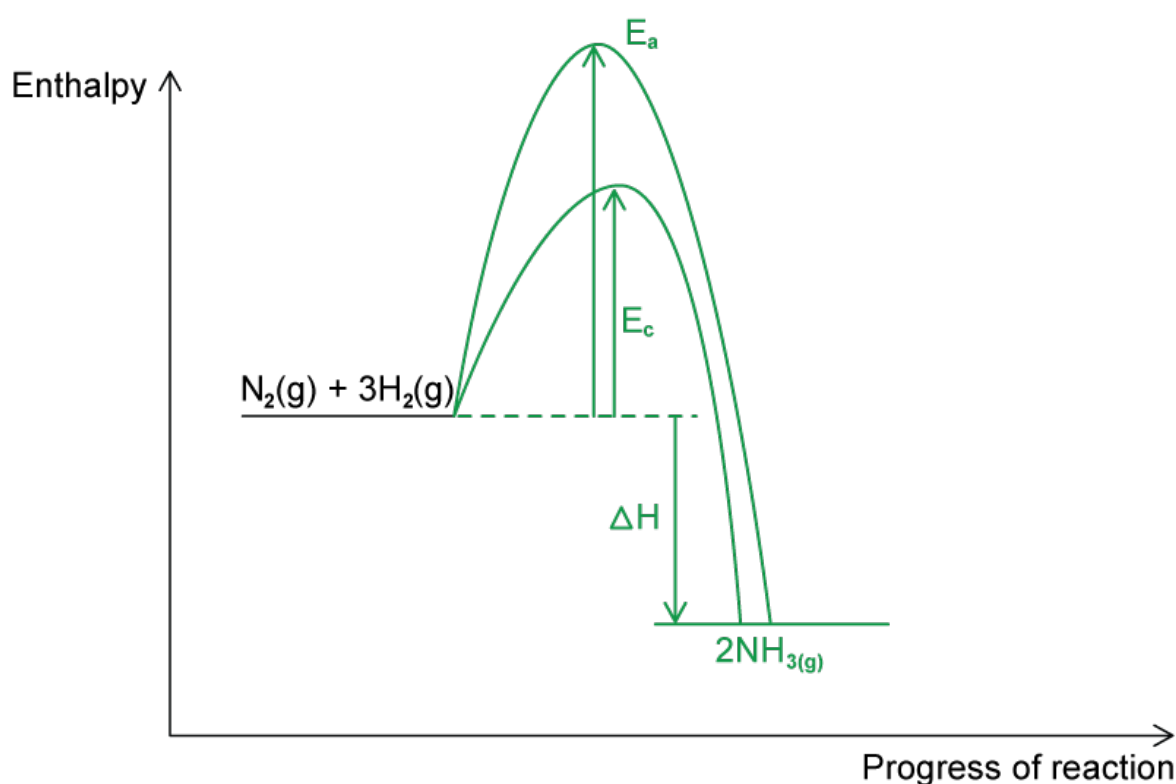
[Total: 2 marks]

- Whilst the effect of surface area on rate of reaction is not specifically covered within this specification, you may have covered it in your previous studies

- The 'suggest' command word indicates that you need to apply your knowledge
- The rate of reaction depends on the frequency of effective collisions
- Larger particles will have a smaller surface area than the same mass of smaller particles, so there is less contact between reactant molecules
- This means fewer collisions will occur, resulting in less frequent effective collisions and the rate of reaction will decrease

5a

a) The completed reaction pathway diagram:



- Products correctly labelled
AND
Products lower than the reactants; [1 mark]
- The enthalpy change of reaction, ΔH , correctly labelled with arrow from the reactants to the products; [1 mark]
- The activation energy of the forward, uncatalysed reaction, E_a , correctly labelled with arrow from the reactants to the peak of the curve; [1 mark]
- The activation energy of the forward, catalysed reaction, E_c , correctly labelled with arrow from the reactants to the peak of the curve

AND

The peak of the curve of the catalysed reaction is lower than the peak of the uncatalysed reaction; [1 mark]

[Total: 4 marks]

- Reaction pathway diagrams need to be drawn carefully
- Ensure that you use single-headed arrows as double-headed arrows will be penalised
- The enthalpy change should be an arrow from the reactants to the products
- Activation energy is shown by an arrow from the reactant to the peak of the curve
- Always use a ruler to draw the arrows
- If you draw an endothermic reaction instead, i.e. the reactants lower than the products, you will lose the 1st mark but could gain the remaining 3 marks

5b

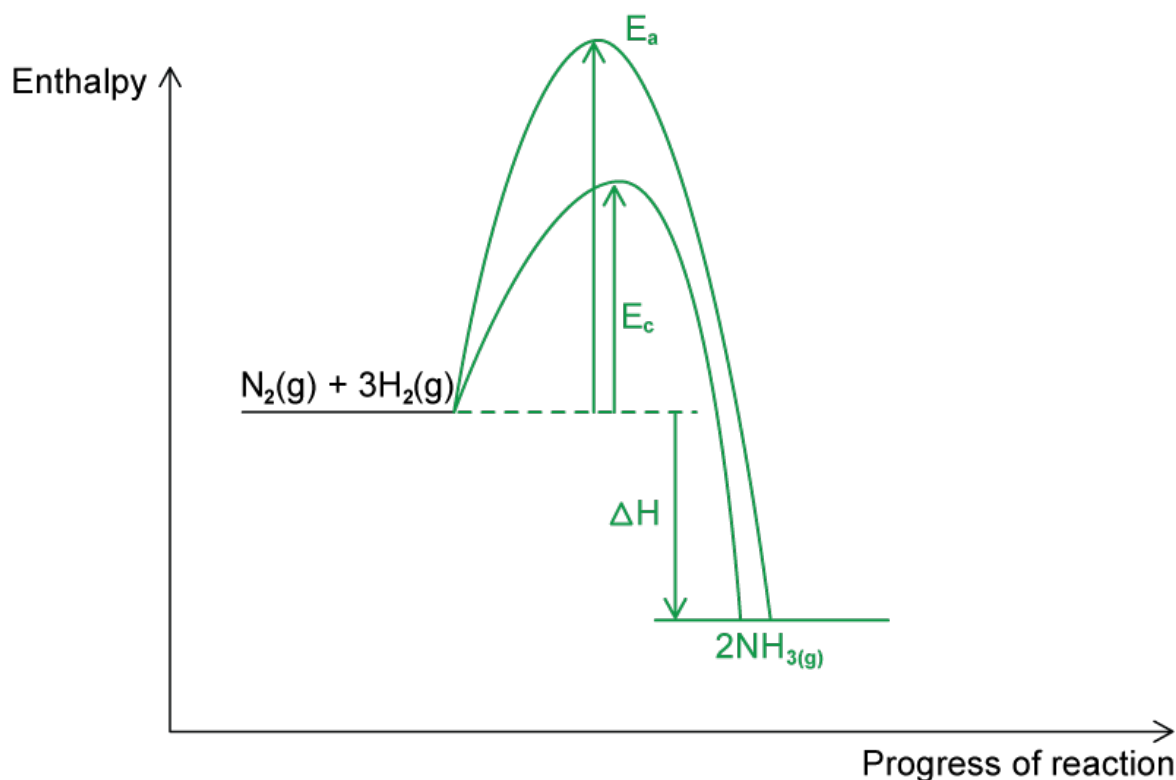
b) The activation energy of the catalysed decomposition of ammonia into nitrogen and hydrogen is:

- $(+92 + 109) = (+)201 \text{ (kJ mol}^{-1}\text{)}$; [1 mark]

[Total: 1 mark]

- The catalysed decomposition of ammonia into nitrogen and hydrogen is just the reverse of the reaction given in part (a)
- So the activation energy of the reverse reaction = $-\Delta H + E_c$ of forward reaction, which is shown on the reaction profile as the energy from the products to the peak of the catalyst reaction curve:





- Products correctly labelled

AND

Products lower than the reactants; [1 mark]

- The enthalpy change of reaction, ΔH , correctly labelled with arrow from the reactants to the products; [1 mark]
- The activation energy of the forward, uncatalysed reaction, E_a , correctly labelled with arrow from the reactants to the peak of the curve; [1 mark]
- The activation energy of the forward, catalysed reaction, E_c , correctly labelled with arrow from the reactants to the peak of the curve

AND

The peak of the curve of the catalysed reaction is lower than the peak of the uncatalysed reaction; [1 mark]

[Total: 4 marks]

- Reaction pathway diagrams need to be drawn carefully
- Ensure that you use single-headed arrows as double-headed arrows will be penalised
- The enthalpy change should be an arrow from the reactants to the products
- Activation energy is shown by an arrow from the reactant to the peak of the curve
- Always use a ruler to draw the arrows

- If you draw an endothermic reaction instead, i.e. the reactants lower than the products, you will lose the 1st mark but could gain the remaining 3 marks

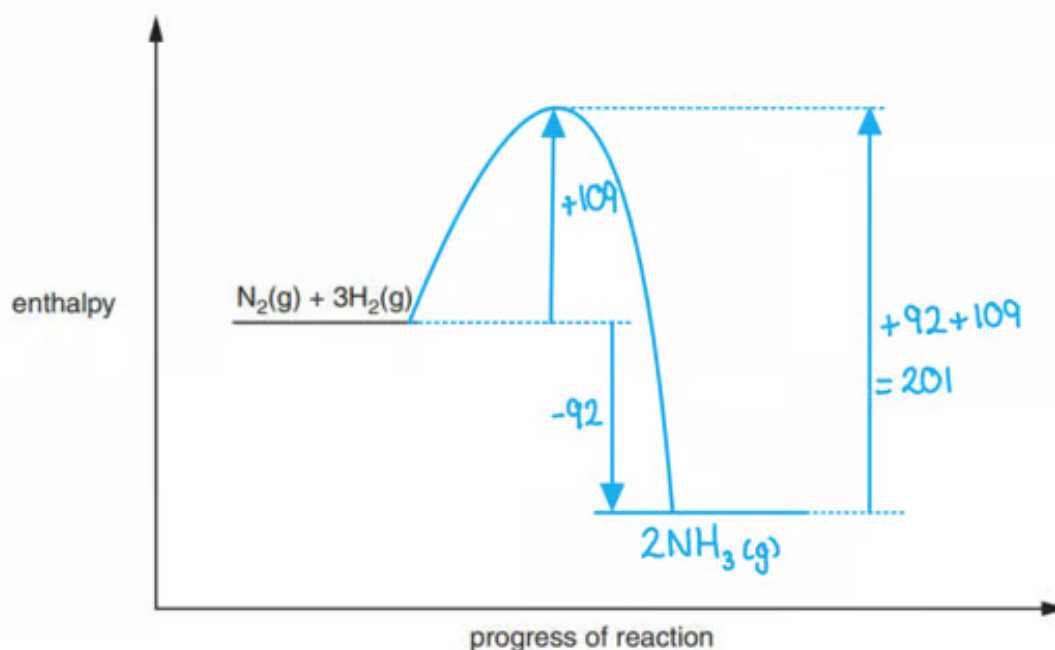
5b

b) The activation energy of the catalysed decomposition of ammonia into nitrogen and hydrogen is:

- $(+92 + 109) = (+)201 \text{ (kJ mol}^{-1}\text{)}$; [1 mark]

[Total: 1 mark]

- The catalysed decomposition of ammonia into nitrogen and hydrogen is just the reverse of the reaction given in part (a)
- So the activation energy of the reverse reaction = $-\Delta H + E_c$ of forward reaction, which is shown on the reaction profile as the energy from the products to the peak of the catalyst reaction curve:



5c

c) Catalysts, such as iron used in the production of ammonia, increase the rate of reaction because:

- Correct drawing of Boltzmann distribution

AND

Axes labelled (number of) molecules and (molecular) energy; [1 mark]

- More molecules with energy above E_a with a catalyst

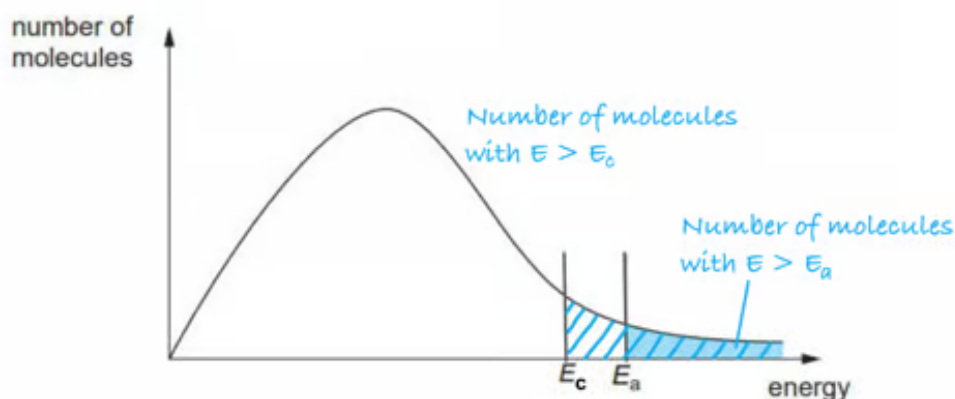
OR

More molecules overcome E_a ; [1 mark]

- More frequent effective / successful collisions; [1 mark]
- Idea that E_a is lowered; [1 mark]
- (Because) the catalyst has a different reaction mechanism / profile; [1 mark]

[Total: 5 marks]

- You can gain the 2nd and 5th mark from the Boltzmann distribution:



- You can use the term particles (but **not** atoms) instead of molecules throughout your answer
- Make sure that your curve starts at the origin and does not touch the x-axis

5d

d)

i) The type of catalyst that platinum is in a catalytic converter is:

- Heterogeneous; [1 mark]

ii) To explain if nitrogen is being oxidised or reduced in this reaction:

- (Nitrogen is) reduced; [1 mark]
- As is oxidation number decreases from +2 (in NO) to 0 (in N₂); [1 mark]

[Total: 3 marks]

Platform for teachers and students to share resources and discuss questions